



AlleyPin

AlleyPin: Enabling convenient, patient-centric healthcare services across Taiwan

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GOOGLE CLOUD RESULTS

- ✓ Reduces 50 percent of cloud infrastructure maintenance workloads
- ✓ Helps accelerate new feature deployment by 3x with [GKE](#)
- ✓ Prevents technical errors with automated alerts sent through [Cloud Monitoring](#)
- ✓ Increases clinic appointment number by 22 percent through [Reserve with Google](#)

Leveraging the cloud infrastructure of Google Cloud to run its healthcare service systems, AlleyPin is able to reduce its IT maintenance workloads by 50 percent while continuously optimizing

its services and operations with data analytics and diverse solutions in the Google ecosystem.

Amidst Taiwan's tech-savvy population, most hospitals and clinics provide an online appointment service. However, due to tight resources, the features of their online appointment systems can be rather limited to create a truly convenient clinic visit experience for their patients.

With that in mind, [AlleyPin](#) is dedicated to helping healthcare professionals provide convenient, patient-centric services through digital technologies. Founded in 2017, the Taiwanese software provider currently offers an online medical appointment platform named PinMed, a patient relationship management system called 1.Talk, as well as healthcare brand consultancy services. It has served more than 1,200 clinics with four million users in Taiwan and counting.



"Our mission is to empower healthcare service providers to enhance their operational efficiency and service quality by offering them comprehensive digital solutions that cover the whole patient journey from pre-visit planning to post-visit care," explains Bill Huang, senior backend engineer at AlleyPin. "Patients can also easily find and make appointments with the healthcare professionals they need on our platform."

Back in 2017, AlleyPin only operated a patient feedback system, which was deployed in a public cloud environment using virtual

machines (VMs). Since the company only had two engineers at that time, it wanted to replace VMs with Kubernetes clusters to reduce its maintenance workloads and invest more resources in product development. With a fully-managed Kubernetes service, Google Cloud became an obvious choice for AlleyPin.

“Google Cloud was the best option because it was the first in the industry to offer a fully managed Kubernetes service. As the only cloud provider with a data center in Taiwan at the time, it also supported faster connectivity and could help us ensure data security.”

Bill Huang

Senior backend engineer, AlleyPin

“We were a small startup with limited resources, so we needed to use the most cost-effective cloud infrastructure to quickly expand our products. Google Cloud was the best option because it was the first in the industry to offer a fully-managed Kubernetes service. As the only cloud provider with a data center in Taiwan at the time, it also supported faster connectivity and could help us ensure data security,” notes Huang.



Enhancing deployment and operational efficiency with fully-managed tools

After the migration of its patient feedback system, AlleyPin started deploying all its new products in Google Cloud. Today, 80 percent of its services are hosted by Kubernetes clusters in Google Kubernetes Engine (GKE), with some internal systems supported by Cloud Run. It also leverages Cloud SQL to run the databases of its services, and Pub/Sub to enable communication between different services, such as automating the sending of review invite messages to patients who went to medical appointments made through AlleyPin’s platform.

With the fully managed tools on Google Cloud, AlleyPin has reduced its maintenance workloads by 50 percent. Furthermore, since GKE enables automated cluster configuration and service isolation, AlleyPin can quickly integrate its healthcare service system with third-party tools and launch a new feature within one week. Without GKE, the team would need to spend two to three weeks to manually configure the deployment environment.

“With GKE, we have greatly enhanced our deployment and operational efficiency, so that we can concentrate on development work and link our systems with more third-party platforms to create better user experiences.”

Jerry Huang

Senior backend engineer, AlleyPin

“With GKE, we have greatly enhanced our deployment and operational efficiency, so that we can concentrate on development work and link our systems with more third-party platforms to create better user experiences,” says Jerry Huang, senior backend engineer at AlleyPin. “For example, GKE supports automated Kubernetes version updates and can flag the resources that block an update when it fails. We would have spent much more time on installing updates if we had run Kubernetes clusters ourselves.”



Enabling secure, zero-downtime digital healthcare services

Google Cloud partner iKala has been assisting AlleyPin’s use of Google Cloud by offering billing services and running security checks. Based on the security review report made by iKala and the Google Cloud team, AlleyPin is able to quickly identify the potential risks and take necessary actions to ensure full security of its appointment platform and patient relationship management system.

Since the migration to Google Cloud, AlleyPin has never experienced any service downtime, partly thanks to the high stability of the Google Cloud infrastructure. To strengthen the reliability of its services, the AlleyPin team employs Cloud Monitoring to set up policies that automatically send alerts when its resource usage reaches a certain level. This way, the team is able to react before a technical error occurs, usually one to two hours in advance, to prevent any system outage.



Facilitating medical appointment making with the Google ecosystem

For AlleyPin, one great advantage of migrating to Google Cloud is that it can easily use other solutions in the Google ecosystem to improve its services. The software provider leverages [Places API](#) in [Google Maps Platform](#) to aggregate clinics' addresses and ratings on Google Maps for users to search on its appointment platform. By sending its clinic customers' data and appointment links to [Reserve with Google](#), it also allows patients to directly make appointments on Google Search and Google Maps. Following the launch of this feature, its clinic customers have seen an average of 22 percent more appointments.

The high compatibility between tools in the Google ecosystem also allows AlleyPin and its clinic customers to effectively optimize their services. By retrieving appointment and review data from Reserve with Google and Google Maps Platform, and having them analyzed in [BigQuery](#), which is then connected with [Looker Studio](#) for visualization, AlleyPin is able to know how to improve its platform to provide more convenient and useful appointment making services. Its clinic customers can also see patients' feedback summarized in one place.

“The diverse solutions in the Google ecosystem have enabled us to offer more accessible and convenient medical appointment services with little development effort.”

Bill Huang

Senior backend engineer, AlleyPin

“The diverse solutions in the Google ecosystem have enabled us to offer more accessible and convenient medical appointment services with little development effort,” says Huang. “We can even collect data from these tools and generate business insights in Google Cloud for product improvement.”



Maximizing productivity and service convenience with AI

Next, AlleyPin is exploring ways to leverage the latest artificial intelligence (AI) technologies to further improve its IT team’s productivity and the convenience of its services. Some of its engineers are testing [Gemini Code Assistant](#) to write codes more quickly, and it plans to use [AlloyDB AI](#) to improve the user experience of AlleyPin’s customer service chatbot and internal search engine. Since AlloyDB AI can support vector data processing without being connected with vector databases, it will help AlleyPin reduce development costs.

“Using the fully managed infrastructure tools of Google Cloud and other Google solutions, we’ve successfully launched reliable and convenient services that benefit both healthcare professionals and patients,” notes Jerry Huang. “We’re now ready to leverage the AI capabilities of Google Cloud to bring our services to the next level.”

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Founded in 2017, **AlleyPin** is a Taiwanese healthcare service software provider dedicated to helping hospitals and clinics embark on digital transformation. Offering a patient relationship management system, an online medical appointment platform and healthcare brand consultancy services, AlleyPin has served 1,200+ clinics and has a user base of more than four million.

Industry: Healthcare and Life Sciences

Location: Taiwan

Products: [Google Kubernetes Engine](#), [BigQuery](#), [Cloud Monitoring](#), [Cloud Run](#), [Cloud SQL](#), [Looker Studio](#), [Pub/Sub](#)

About Google Cloud partner—[iKala](#)

[iKala](#) is a Taiwanese digitalization and AI commerce solution provider founded in 2011. Its subsidiary iKala Cloud is a Google Cloud partner offering consultation services for cloud infrastructure, hybrid cloud solutions, machine learning, and marketing applications to help enterprises overcome challenges brought by digital transformation. The company has served more than 400 enterprise customers in Asia Pacific for their use of Google Cloud.

GOOGLE CLOUD PARTNERS

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